Concussion Symptom Cutoffs for Identification and Prognosis of Sports-Related Concussion

Role of Time Since Injury

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Background: Symptom reporting with scales such as the Post-Concussion Symptom Scale (PCSS) is one of the most sensitive markers of concussed status and/or recovery time, It is known that time from injury until initial clinic visit affects symptom presentation and recovery outcomes, but no study to date has evaluated changes in clinical cutoff scores for the PCSS based on earlier versus later clinical presentation postconcussion.

Purpose: To evaluate if time since injury after sports-related concussion (SRC) affects clinical cutoff scores for total PCSS and PCSS factors in differentiating athletes with SRC from healthy controls and predicting prolonged recovery (>30 days) after SRC.

Study Design: Cohort study; Level of evidence, 3.

Methods: A chart review was conducted of clinical data from patients with SRC (age, 13-25 years; n = 588; female, n = 299) who presented to concussion specialty clinics. Participants were categorized on the basis of time from injury: early (≤7 days; n = 348) and late (8-21 days; n = 240). Outcomes were total symptom severity (ie, total PCSS score) and total score for each of 4 symptom factors (cognitive/migraine/fatigue [CMF], affective, sleep, and somatic). Area under the curve (AUC) analyses were conducted using the Youden index to optimize sensitivity and specificity cutoffs.

Results: In the early group, the CMF factor (cutoff, ≥7; AUC = 0.944), affective factor (cutoff, ≥1; AUC = 0.614), and total PCSS (cutoff, ≥7; AUC = 0.889) differentiated athletes with SRC from controls. In the late group, the CMF factor cutoff was reduced (cutoff, ≥4; AUC = 0.945), while the total PCSS score (cutoff, ≥7; AUC = 0.892), affective factor (cutoff, ≥1; AUC = 0.603), and sleep factor (cutoff, ≥1; AUC = 0.609) remained the same. In the early cohort, the CMF factor was the strongest predictor of protracted recovery (cutoff, ≥23; AUC = 0.717), followed by the total PCSS (cutoff, ≥39; AUC = 0.695) and affective factor (cutoff, ≥2; AUC = 0.614). The affective factor (cutoff, ≥1; AUC = 0.642) and total PCSS (cutoff, ≥35; AUC = 0.592) were significant predictors in the late cohort, but the cutoff threshold was reduced.

Conclusion: The findings indicate that PCSS symptom clinical cutoffs for identifying injury and recovery prognosis change on the basis of time since injury. Specifically, the combination of CMF, affective, and sleep factors is the best differentiator of athletes with SRC from controls regardless of time since injury. Furthermore, the CMF factor is the most robust predictor of prolonged recovery if the patient is within 1 week of SRC, whereas the affective factor is the most robust predictor of prolonged recovery if the patient is within 2 to 3 weeks of SRC.

Keywords: concussion; symptoms; PCSS; clinical cutoffs; recovery

Sports-related concussion (SRC) remains a public health concern, as up to 3.8 million occur in the United States annually.[1] According to the most recent international consensus statement, concussion symptom reporting provides the most sensitive marker of concussed status.[1] Given its ease of implementation and high information yield, the Post-Concussion Symptom Scale (PCSS) is one of the most common clinical tools used to evaluate concussion-specific symptom severity.[8] The PCSS is a 22-item self-report scale in which patients rate symptom severity from 0 to 6. The PCSS can be administered at any time throughout recovery from concussion, as a measure to track changes in symptom burden. While total symptom severity has demonstrated clinical utility in prediction of

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prolonged recovery,[10-12] clinicians have recognized that individual symptoms or subgroups of related symptoms within the PCSS may provide more specific information about the individual's unique symptom burden.[5]

Subsequently, researchers have identified distinct symptom factors within PCSS, ranging from 3 to 5 depending on methodological differences.[2,4,5,13] Pardini et al[13] first reported an exploratory factor analysis on the PCSS in athletes, with 4 distinct domains: cognitive, migraine, emotional, and sleep. Lau et al[7] subsequently reported cutoff scores for these domains at predicting protracted (>14 days) recovery in high school football athletes with SRC within the previous 3 days. Specifically, the authors reported that cognitive (cutoff, ≥18) and migraine (cutoff, ≥15) significantly predicted SRC, while emotional and sleep did not. Kontos et al[5] extended the work of Pardini et al by demonstrating that cognitive, migraine, and fatigue symptoms formed a "global" concussion factor with 3 additional factors (affective, somatic, and sleep) within 1 to 7 days from injury.

However, not all athletes with recent SRC present for initial evaluation within 7 days. Time (ie, number of days) from concussion to first clinical visit is a significant predictor of normal concussion recovery time and may play a role in symptom presentation.[6] Specifically, time from injury may affect which symptom factors are most prominent, such as the global factor within the first week of injury.[5] However, no research to date has investigated the role of time from injury on concussion symptom presentation cutoffs. Furthermore, while total PCSS score has been utilized for concussion identification and recovery prediction after concussion, no clinical cutoffs have been reported with sensitivity and specificity values.[10-12] Therefore, the purpose of this study was 2-fold: the primary purpose was to evaluate if time from SRC (early presentation, ≤7 days; late presentation, 8 to 21 days) affects clinical cutoff scores for total PCSS and individual PCSS factors in differentiating athletes with SRC from healthy controls and in predicting prolonged recovery (>30 days) in athletes after SRC. A secondary purpose was to identify the most robust individual PCSS factors for differentiating athletes with SRC from healthy controls and in predicting prolonged recovery (>30 days) in athletes after SRC.

METHODS

Design and Participants

The present study was a cross-sectional chart review of clinical and demographic data from ongoing research on patients with SRCs (age, 13-25 years) who presented to a concussion specialty clinic or participated in seasonal baseline testing in the Midwest and Central Midwest regions of the United States between September 2011 and February 2019 (n = 588; female, n = 319). Athletes were eligible to participate in research if they had a diagnosed symptomatic SRC per current international consensus ≤21 days from injury.[9] Athletes with moderate to severe traumatic brain injury, neurological disorder, or pre-existing vestibular disorder were excluded from analysis. The healthy control group was recruited and tested using identical procedures and inclusion/exclusion criteria, except for no recent SRC. Eligible participants in the control group were recruited into the study by research staff during seasonal baseline testing. Eligible participants from the group with SRC were recruited into the study by research staff if they met inclusion criteria during their first clinic visit.

Measures

Patient Characteristics and Medical History. Athletes reported demographics, relevant medical history, and injury information in a standardized clinical interview. Characteristics included age, sex, and sport. Medical history outcomes included concussion history, as well as history of attention-deficit/hyperactivity disorder (ADHD) / learning disability (LD), migraine/headaches, and history of motion sickness. Participants were categorized into groups based on time from injury: early (≤7 days) and late (8-21 days), as done previously.[6]

Concussion Symptoms. Participants completed the PCSS, which is a self-report symptom severity survey.[5] The PCSS has 22 items comprising concussion-related symptoms, rated on a 7-point Likert scale from 0 (none) to 6 (severe). Maximum score for the PCSS is 132. The PCSS takes <3 minutes to complete. Scoring methods for the PCSS were total symptom severity (ie, total PCSS score) and total score for each factor (ie, cognitive/migraine/fatigue [CMF], affective, sleep, and somatic), as defined by Kontos et al.[5] The CMF factor included items such as headache, dizziness, fatigue, drowsiness, sensitivity to light/noise, feeling slowed down, mentally foggy, difficulty concentrating, and difficulty remembering. The affective factor included sadness, nervousness, and feeling more emotional. The sleep factor included trouble falling asleep and sleeping less than usual. The somatic factor included vomiting and numbness.

Recovery Time. Recovery was operationally defined by current consensus guidelines; clearance for full activity

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TABLE 1
Differences in Medical History Between Controls and Athletes With Concussion Who Presented to the Clinic Early (≤7 Days) and Late (8-21 Days)[a]

	Participants, Mean ± SD or No. (%)		
	Controls (n = 44)	Early (n = 348)	Late (n = 240)
Age, y	15.5 ± 1.9	15.4 ± 1.9	15.6 ± 2.0
Days			
From injury to first visit	—	4.0 ± 1.7[b]	13.9 ± 5.7[c]
Until recovery	—	29.7 ± 32.3[b]	44.2 ± 34.1[c]
From first visit to recovery	—	25.8 ± 32.3	30.1 ± 32.9
Concussion history	0.1 ± 0.3[b]	0.6 ± 0.8[c]	0.4 ± 0.7[c]
Female sex	20 (45)	166 (48)	133 (55)
ADHD/LD	4 (9)[b]	7 (2)[c]	3 (1)[c]
Migraine/headache	3 (7)[b]	120 (34)[c]	93 (39)[c]
Motion sickness	9 (20)	98 (28)	64 (27)
Total PCSS	5.4 ± 10.3[b]	29.0 ± 19.9[c]	27.1 ± 18.2[c]
Factor			
CMF	1.7 ± 3.3[b]	18.9 ± 11.7[c]	17.6 ± 10.8[c]
Somatic	0.2 ± 1.2	0.4 ± 1.1	0.4 ± 1.0
Affective	1.3 ± 2.6	2.3 ± 3.4	2.2 ± 3.3
Sleep	1.3 ± 2.3[b]	1.7 ± 2.4[c]	2.2 ± 2.8[b]

[a]Dashes indicate data not available. ADHD/LD, attention-deficit/hyperactivity disorder/learning disability; CMF, cognitive/migraine/fatigue; PCSS, Post-Concussion Symptom Scale.
[b,c]Dissimilar letters denote statistical differences ($P < .05$).

was indicated after a return to baseline level of symptoms, including cognitive, ocular, and vestibular performance at rest, with no increased symptoms after exertion.[9] Recovery time was recorded as the number of days from the date of injury to the date of clearance for full activity. Because most athletes recover within 4 weeks, according to the aforementioned definition, participants were separated into 2 cohorts for analysis: ≤30 days and >30 days.[18]

Procedures

Athletes were informed about study procedures before providing written informed consent. If athletes were <18 years of age, child assent was obtained as well as parent informed consent before study participation. After consent, athletes completed demographics, injury information, and PCSS surveys in the initial portion of a typical clinical visit. This study was approved by the University of Pittsburgh and University of Arkansas institutional review boards.

Statistical Analysis

Differences between healthy controls and athletes who were concussed in demographics and relevant medical history were evaluated using 1-way analysis of variance for continuous variables and chi-square testing for categorical variables. Continuous data were assessed for normality using Shapiro-Wilk tests before analysis. Receiver operating characteristic (ROC) curves, with area under the curve (AUC) analyses, were conducted to determine optimal cutoff scores for PCSS total and cluster scores. To maximize the accuracy of differentiating athletes with SRC from healthy controls and to identify athletes with SRC and prolonged recovery, a cutoff score was identified with the Youden index (J). The Youden index is determined by identifying the maximum score per the following formula: J = (sensitivity + specificity – 1). Two multivariate forward stepwise logistic regressions were conducted to identify which symptom factors (CMF, affective, somatic, sleep) were the most robust differentiators between athletes with SRC and controls and predictors of prolonged recovery in each SRC group (early and late). If multiple symptom factors were retained in the model, ROC curves with AUC analyses were conducted again to determine if the combination of included symptom factors was more predictive than were individual symptom factors. Significance was set at $P < .05$ a priori. Analyses were conducted with SPSS software (v 26; IBM Corp).

RESULTS

Patient characteristics and medical history are reported in Table 1. Participants with complete recovery data were included for the predictive utility analysis (early, n = 270, 78%; late, n = 180, 75%). The early group had 79 (29.3%) participants with prolonged recovery (>30 days), while the late group had 106 (58.9%). Both groups reported significantly more previous concussions than controls. Both groups also reported a higher rate of premorbid history of headache/migraine (early, 34%; late, 39%) than controls (7%), and controls reported a higher rate of ADHD/LD (9% vs 2% and 1%, respectively). There were no differences between healthy controls and athletes with SRC in age, number of female participants, and number of athletes

TABLE 2
Cutoffs for Identifying SRC From Healthy Controls Stratified by Early ($\leq$7 Days) or Late (8-21 Days) Clinical Presentation[a]

	Cutoff	SN	SP	J[b]	AUC	*P* Value
$\leq$7 d (n = 348)						
PCSS total	$\geq$7	0.88	0.80	0.67	0.889	<.001[c]
Factor						
CMF	$\geq$7	0.85	0.91	0.76	0.944	<.001[c]
Affective	$\geq$1	0.54	0.68	0.22	0.614	.013[c]
Sleep	$\geq$1	0.50	0.98	0.17	0.567	.149
Somatic	$\geq$1	0.19	0.98	0.17	0.581	.079
8-21 d (n = 240)						
PCSS total	$\geq$7	0.87	0.80	0.66	0.892	<.001[c]
Factor						
CMF	$\geq$4	0.90	0.84	0.75	0.945	<.001[c]
Affective	$\geq$1	0.51	0.68	0.20	0.603	.030[c]
Sleep	$\geq$1	0.56	0.66	0.22	0.609	.022[c]
Somatic	$\geq$1	0.19	0.98	0.17	0.580	.091

[a]AUC, area under the curve; CMF, cognitive/migraine/fatigue; PCSS, Post-Concussion Symptom Scale; SN, sensitivity; SP, specificity; SRC, sports-related concussion.
[b]The Youden index is determined by identifying the maximum score per the following formula: J = (sensitivity + specificity – 1).
[c]$P < .05$.

with premorbid motion sickness history. There were no differences between the early and late groups in the somatic or affective factor. There were no differences between groups in total PCSS score or any of the PCSS factors, with the exception of the late group reporting a higher sleep score than the early group (+ 0.5).

Utility of PCSS Factors for Identifying Concussed Athletes From Controls

Optimal cutoff values, sensitivity/specificity, Youden index, and AUC for each PCSS outcome at identifying concussion are reported in Table 2. ROC curves are displayed in Figure 1. In the early group, a CMF factor score $\geq$7 identified athletes who were concussed from controls with 94.4% accuracy (AUC = 0.944). Additionally, total PCSS score (cutoff, $\geq$7; AUC = 0.889) and the affective factor (cutoff, $\geq$1; AUC = 0.614) significantly differentiated between athletes who were concussed and controls. Regarding regression analysis, CMF ($P < .001$), affective ($P < .001$), and sleep ($P < .001$) were the factors included in the final regression model as the most robust differentiators between athletes with SRC and controls. The final model had an AUC of 0.980 ($P < .001$) (Figure 2).

In the late group, a CMF factor score $\geq$4 identified athletes who were concussed from controls with 93.5% accuracy (AUC = 0.945). Additionally, total PCSS score (cutoff, $\geq$7; AUC = 0.892) and the affective (cutoff, 1; AUC = 0.603) and sleep (cutoff, $\geq$1; AUC = 0.609) factors significantly differentiated athletes with SRC from controls. Regarding regression analysis, CMF ($P < .001$), affective ($P = .045$), and sleep ($P = .002$) factors were included in the final regression model as the most robust differentiators between athletes with SRC and controls. The final model had an AUC of 0.963 ($P < .001$) (Figure 2).

Utility of PCSS Factors for Predicting Protracted Recovery From SRC

Optimal cutoff values, sensitivity/specificity, Youden index, and AUC for each PCSS outcome at identifying prolonged recovery are reported in Table 3. ROC curves are displayed in Figure 3. In the early group, total PCSS, as well as the CMF and affective factors, significantly predicted recovery >30 days. CMF factor score $\geq$23 predicted prolonged recovery at 71.7% accuracy (AUC = 0.717). Total PCSS (cutoff, $\geq$39; AUC = 0.695) and the affective factor (cutoff, $\geq$2; AUC = 0.614) also significantly predicted prolonged recovery. Regarding regression analysis, only CMF was included in the final model as a predictor of prolonged recovery (Figure 4).

In the late group, total PCSS and affective factor significantly predicted recovery >30 days (Table 3). Affective factor score $\geq$1 predicted prolonged recovery with 64.2% accuracy (AUC = 0.642), followed by total PCSS (cutoff, $\geq$35; AUC = 0.592). Regarding regression analysis, only the affective factor was included in the final model as a predictor of prolonged recovery (Figure 4).

DISCUSSION

The purpose of this study was to evaluate the potential effect of time from injury on the identifying and predictive utility of total PCSS and PCSS symptom factors in athletes with SRC. The primary finding of this study is that PCSS symptom factors more accurately differentiate athletes with SRC from controls and predict prolonged recovery than does the total PCSS score, regardless of time since injury. Specifically, the CMF, affective, and sleep factors combined to differentiate athletes with SRC from controls at 98% accuracy in the early group and 96% accuracy in

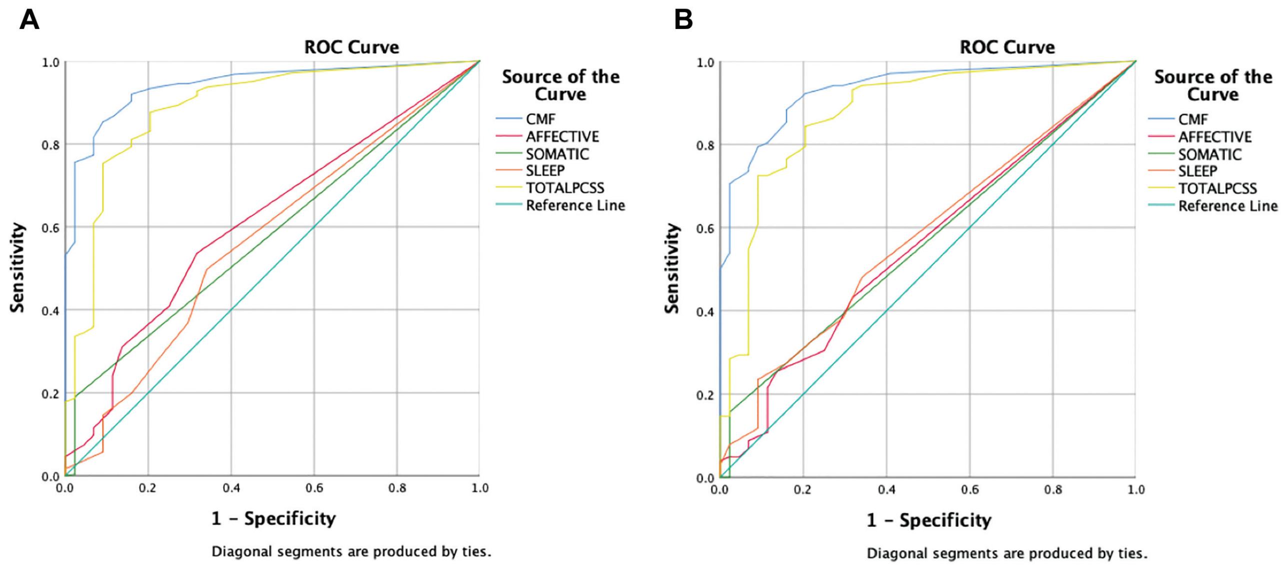

Figure 1. ROC curve results for total PCSS score and PCSS symptom factors to differentiate between healthy controls and athletes with suspected SRC whose initial visit occurred within (A) 1 to 7 days after injury and (B) 8 to 21 days after injury. CMF, cognitive/migraine/fatigue; PCSS, Post-Concussion Symptom Scale; ROC, receiver operating characteristic; SRC, sports-related concussion.

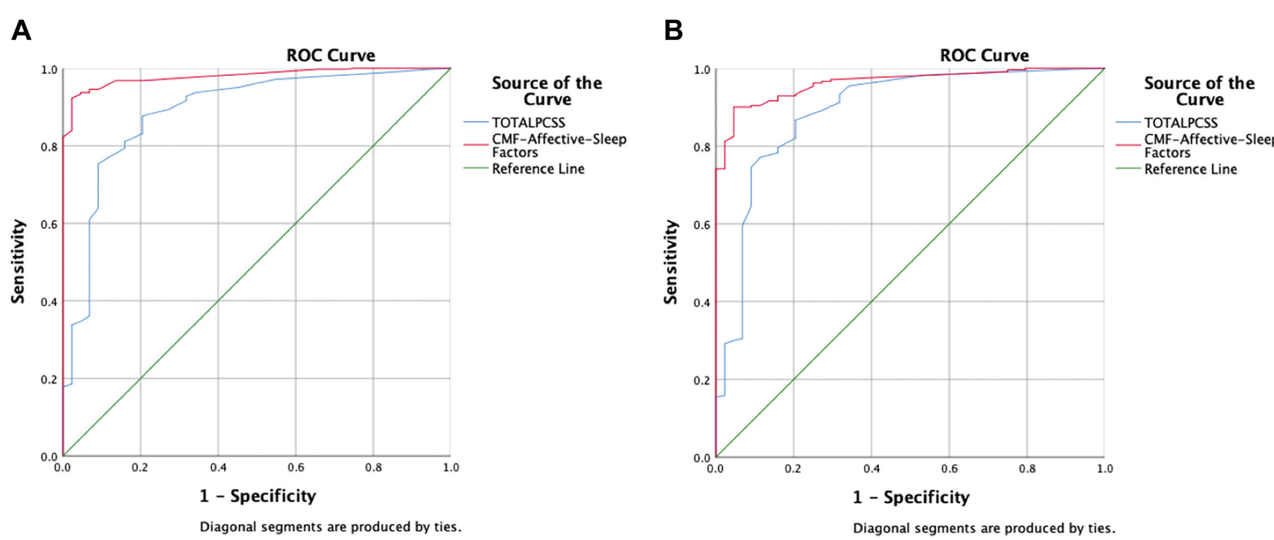

Figure 2. ROC curve results for total PCSS score and the combined CMF-affective-sleep symptom factor to differentiate between healthy controls and athletes with suspected SRC whose initial visit occurred within (A) 1 to 7 days after injury and (B) within 8 to 21 days after injury (bottom image). CMF, cognitive/migraine/fatigue; PCSS, Post-Concussion Symptom Scale; ROC, receiver operating characteristic; SRC, sports-related concussion.

the late group. These identification accuracies were higher than any individual factor and the total PCSS score (Table 2). Furthermore, CMF was the only included symptom factor in the final regression model to predict prolonged recovery in the early group, while the affective factor was the only included symptom factor in the final regression model to predict prolonged recovery in the late group. In both cases, these individual factors predicted prolonged recovery with higher accuracy than did the PCSS total score (Table 3). The identifying cutoffs presented here may help inform clinicians on SRC diagnosis when suspected within 1 week or 2 to 3 weeks of injury in an adolescent and young adult population. The predictive cutoffs can potentially be used to inform treatment strategies and management decisions for athletes who present on different timelines from injury and/or with different symptom severity.

TABLE 3
Cutoffs for Predicting Prolonged Recovery From SRC Stratified by Early ($\leq$7 Days) or Late (8-21 Days) Clinical Presentation[a]

	Cutoff	SN	SP	J[b]	AUC	P Value
≤7 d (n = 270)						
PCSS total	≥39	0.52	0.79	0.31	0.695	<.001[c]
Factor						
CMF	≥23	0.58	0.74	0.33	0.717	<.001[c]
Affective	≥2	0.55	0.65	0.19	0.614	.004[c]
Sleep	≥4	0.30	0.84	0.14	0.564	.101
Somatic	≥1	0.27	0.85	0.12	0.558	.135
8-21 d (n = 180)						
PCSS total	≥35	0.37	0.84	0.21	0.592	.035[c]
Factor						
CMF	≥21	0.45	0.72	0.17	0.571	.106
Affective	≥1	0.65	0.59	0.24	0.642	.001[c]
Sleep	≥3	0.42	0.71	0.13	0.579	.071
Somatic	≥1	0.21	0.84	0.05	0.518	.682

[a]AUC, area under the curve; CMF, cognitive/migraine/fatigue; PCSS, Post-Concussion Symptom Scale; SN, sensitivity; SP, specificity; SRC, sports-related concussion.
[b]The Youden index is determined by identifying the maximum score per the following formula: J = (sensitivity + specificity − 1).
[c]$P < .05$.

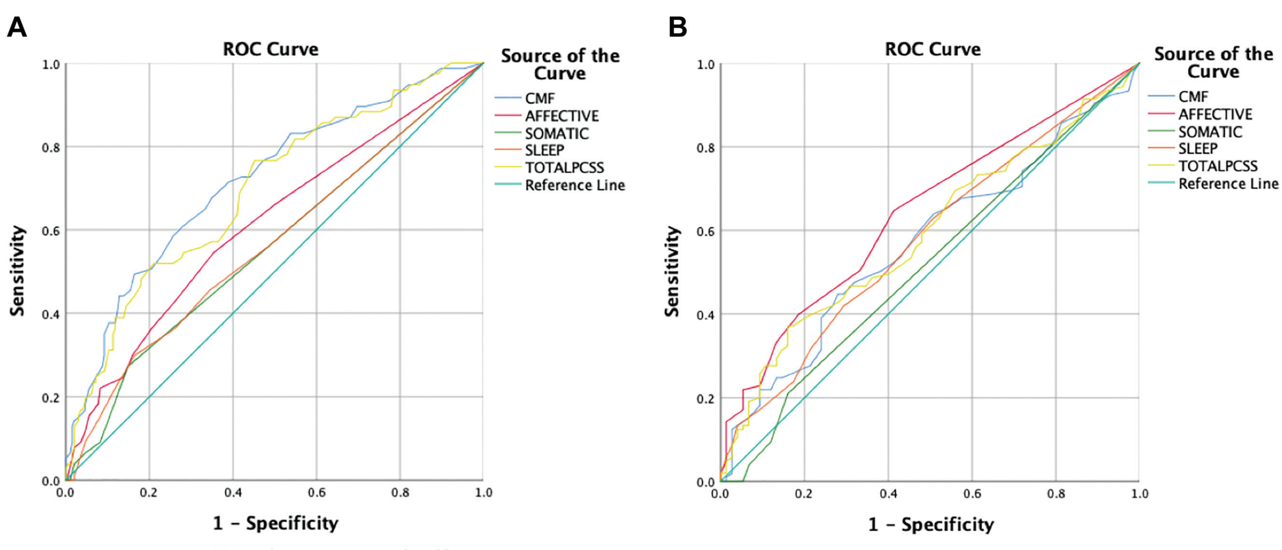

Figure 3. ROC curve results for total PCSS score and PCSS symptom factors to predict recovery >30 days for athletes whose initial visit occurred within (A) 1 to 7 days after injury and within (B) 8 to 21 days after injury. CMF, cognitive/migraine/fatigue; PCSS, Post-Concussion Symptom Scale; ROC, receiver operating characteristic.

Identifying Utility

The result of a total PCSS score ≥7 significantly differentiating athletes with SRC from controls is in agreement with a study by Iverson et al,[3] who reported that 76% of healthy controls had a total PCSS score ≤6. However, the results of this study suggest that the combination of CMF, affective, and sleep factor scores is a better identification tool than total PCSS score in this population, as this combination was the most accurate differentiator between athletes with SRC and controls, both early (98%) and late (96%). Furthermore, CMF alone had a higher accuracy than did total PCSS score for the early and late groups. For the early group, a cutoff ≥7 for the CMF factor had a specificity of 91%, as compared with the total PCSS specificity of 80% (Table 2). For the late group, a cutoff score ≥4 for the CMF factor differentiated athletes with SRC from controls with higher sensitivity (90%) and specificity (84%) than that of the total PCSS score (sensitivity, 87%; specificity, 80%). The reduction in optimal cutoff score for

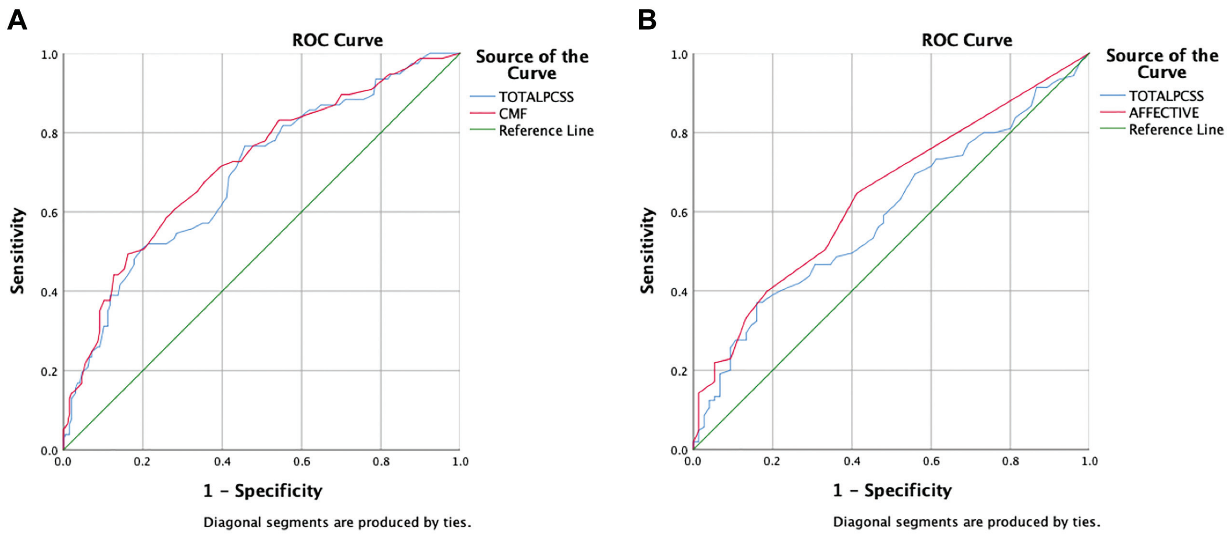

Figure 4. ROC curve results for total PCSS score and the most robust PCSS symptom factor predictor of prolonged recovery (>30 days) for athletes whose initial visit occurred within (A) 1 to 7 days after injury (top image) and within (B) 8 to 21 days after injury (bottom image). CMF, cognitive/migraine/fatigue; PCSS, Post-Concussion Symptom Scale; ROC, receiver operating characteristic.

CMF at the later time point may suggest that CMF encapsulates the "global" symptom presentation of early SRC (eg, headache, phono/photosensitivity, fatigue). Perhaps the lower cutoff score for the late group indicates that CMF symptoms had somewhat abated in the longer period between concussion and clinic presentation. Additionally, reporting any affective or sleep symptoms significantly differentiated between athletes with SRC and controls. This result may provide meaningful information for clinicians who treat an athlete with suspected SRC who presents to the clinic 2 to 3 weeks after injury. Specifically, "typical" symptoms (eg, headache, fatigue) may not be as prominent, and different symptom domains may be more pronounced (eg, sadness, nervousness, sleeping less than usual).

Predictive Utility

The results of this study suggest that the CMF factor and PCSS total score have significant clinical utility in predicting protracted recovery in patients with SRC who present early, whereas the affective factor and PCSS total score have utility in predicting protracted recovery in patients with SRC who present late (Table 3). This result supports the recent consensus that the most sensitive predictor of protracted recovery is symptom severity.[1] Regardless of presentation time, total PCSS scores were predictive of prolonged recovery. Interestingly, the CMF factor was the strongest predictor in the early group, while the affective factor was the strongest predictor in the late group (Figures 2 and 4). This result supports the notion that a "global" response (ie, higher CMF factor scores) is common within the first week after concussion.[5] Lingering symptoms, within the CMF domain or others, past the first week seem to be indicative of a protracted recovery. For the clinician, this study provides objective support for the role of overall symptom burden (PCSS $\geq$35) after SRC in predicting prolonged recovery.

Beyond severity of symptoms, reporting any symptoms within certain domains may be indicative of more complicated recovery after SRC. Within the early cohort, affective symptoms may be problematic, as they were predictive of protracted recovery at a far lesser symptom threshold than were total PCSS or the CMF factor (Table 3). Furthermore, the affective factor was the strongest predictor of protracted recovery in the late cohort, where reporting affective symptoms of any severity (ie, $\geq$1 out of a total possible severity of 18) was predictive of recovery >30 days (Figure 4). Affective symptoms have been demonstrated to increase risk for prolonged recovery after concussion in several populations.[14-17] The results of this study suggest that clinicians may seek to increase awareness regarding the type of symptoms being reported relative to the time scale from injury.

On a broader scale, total symptom severity, as measured by the total PCSS score at initial visit, has repeatedly been associated with prolonged recovery. Meehan and colleagues[11,12] reported increased odds for protracted recovery with higher initial PCSS scores in pediatric[11] and adult[12] populations with SRC. Furthermore, Meehan et al[10] reported that 86% of athletes aged 7 to 26 years had symptom resolution within 28 days of injury with an initial PCSS score <13 (out of 132). While this was an important contribution to understanding the role of initial PCSS score on outcomes, the authors defined recovery as complete symptom resolution (ie, a PCSS score of 0) in these studies. According to the most recent consensus statements, symptom resolution and clinical recovery may not be equal.[1,9] Therefore, symptom resolution in these studies may not have been a "true" marker of recovery from concussion in all cases. The results of the current study provide clinical cutoffs of a valid and reliable symptom survey that can be used by clinicians to identify athletes with SRC and provide a prognosis for recovery time.

This study has limitations. Generalizability is limited to athletes with SRC in an age range from adolescent to young adult (13-25 years). While the study demonstrated differences between athletes who reported to a concussion clinic within a week and those who reported at 2 to 3 weeks, it is unclear what causes those differences. Management strategies before the initial visit were not recorded for analysis. If athletes in the late group adopted a "wait and see" approach to recovery or continued to participate in sports, exercise at high intensity, or isolate themselves from normal activities of daily living, they could have inadvertently prolonged recovery. There is the potential for selection bias, as the early and late groups reported a premorbid history of headache/migraine and controls had higher rates of ADHD/LD. These differences may have affected the symptom factors in the concussed groups. Additionally, athletes with SRC who present earlier to concussion clinics for evaluation may have differences from those who visit concussion clinics later or not at all, such as socioeconomic status, travel availability, insurance, and so on. These factors were not available for analysis in the current study but should be the focus of future work. Future studies should also generate clinical cutoff scores for other populations at high risk of concussion, such as the military, as well as investigate the role of activity/management strategies before clinic presentation.

CONCLUSION

Initial symptom burden is considered the most sensitive identification and prognostic outcome after SRC. However, reliable cutoff scores for the PCSS, one of the most commonly used concussion symptom reports, had not previously been established on the basis of time since injury. The results of this study indicate that the value of certain cutoffs changes according to whether the patient with an SRC is within 1 week or 2 to 3 weeks of recovery. Specifically, the combination of CMF, affective, and sleep factors is the best differentiator of athletes with SRC from controls regardless of time since injury. Furthermore, CMF is the most robust predictor of prolonged recovery if the patient is within 1 week of an SRC, whereas the affective factor is the most robust predictor of prolonged recovery if the patient is within 2 to 3 weeks of SRC. Clinicians can apply the cutoffs presented here to identify an SRC when suspected, in concordance with a multifaceted clinical examination, as well as to guide management and treatment strategies when risk for protracted recovery is indicated.

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